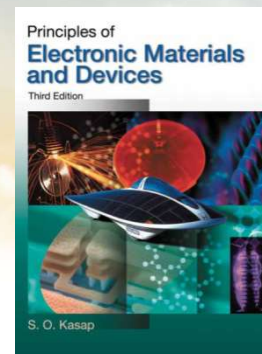




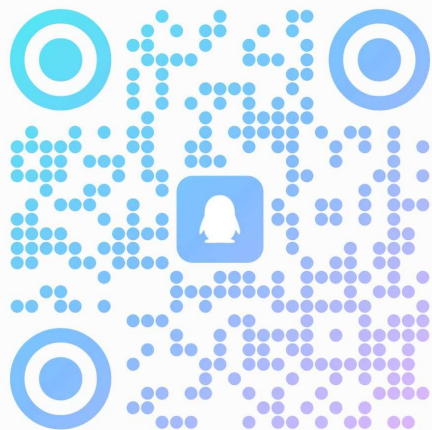
南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

《固态电子学》 Solid-State Electronics



2024 SSE (RUI)

群号: 288257836



Preface
General
Introduction

Dr. Rui CHEN (陈锐)

Email: chenr@sustech.edu.cn

Office: Room 428 at College of
Engineering South Tower

Tel: 0755-88018563



Some slides are modified from other Profs' PPTs. Their contributions are greatly acknowledged.

About the Instructor

Instructor: Dr. Rui CHEN (陈锐)

(every) TUE (1-2), Room 406

(even) THU (1-2), Room 406



Professor

Ph.D @ NTU, Singapore

Ph.D @ Xiamen University

BS @ Xiamen University

Email: chenr@sustech.edu.cn

Through the course, we learn about fundamental physics for electrical/electronic materials and devices, semiconductors. The course is the prerequisite course for many EEE&MSE&ME courses such as semiconductor devices, integrated circuit, lasers, photonics devices, etc.



南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

材料

新颖发光材料的开发

器件

各类光电器件的设计与制备

**光电信息
科学与工程**

物理

光与物质的相互作用



光电专业群



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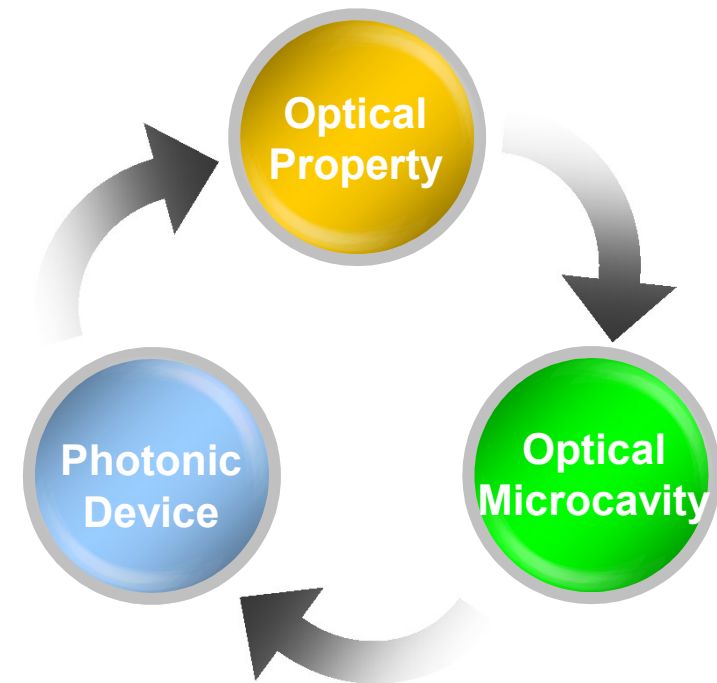
Laser Spectroscopy



- ✓ **Steady State Spectroscopy** (*Absorption, Reflection, Photoluminescence, Photoluminescence Excitation, etc*)
- ✓ **Nonlinear Optics** (*Z-Scan, Multi-Photon Spectroscopy*)
- ✓ **Time-Resolved Spectroscopy** (*Time-Resolved Photoluminescence*)
- ✓ **Ultra-Fast Pump-Probe Transient Absorption Spectroscopy**

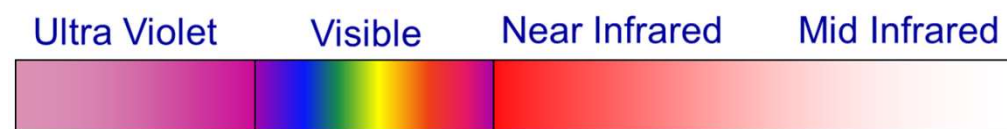
Research Summarize

Optical Property	Investigation of Carrier Dynamic; Excitonic Property; Energy Transfer
Optical Microcavity	Fabry-Pérot Cavity; Random Cavity; Whispering Gallery Modes
Photonic Device	Light Emitting Diode; Temperature Sensor; Refractive Index Sensor



Laser Spectroscopy

<https://eee.sustech.edu.cn/p/chenrui>



PMT



CCD



InGaAs



InSb



Spectrum

Time



Temperature



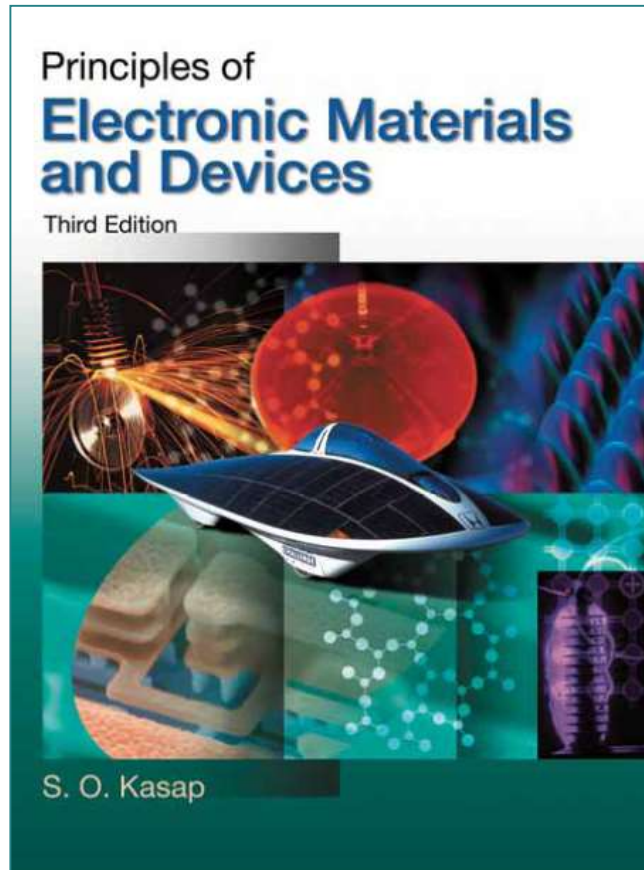
Space



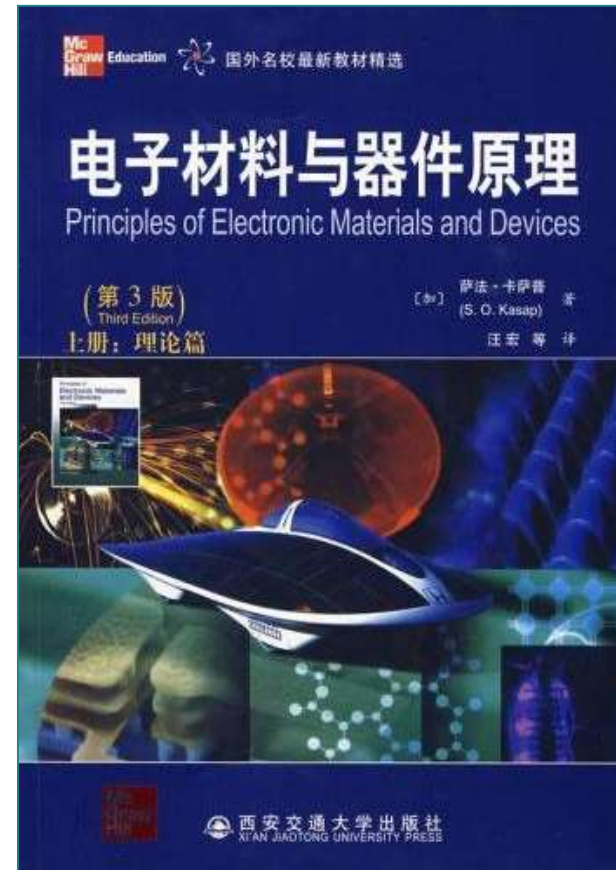
南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

Course References

*Principles of
Electronic Materials and Devices
Third Edition, S.O. Kasap*



电子材料与器件原理
第三版，萨法·卡萨普
汪宏等（译）



南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

Solid-State Electronics

I. Course Information

Course ID: EE203

Name: Solid-State Electronics

Credit: 3

Content: 1) Elementary Materials Science Concepts; 2) Electrical and Thermal Conduction; 3) Elementary Quantum Physics; 4) Modern Theory of Solids.

Target; Persistence; Thinking

目标；坚持；思考

II. Grading

Final score: **attendance 10%, homework 20% , mid-term exam 30%, and final exam 40%**. The mid-term and final examinations are **closed-book**. The questions involve basic concepts, basic principles, analysis and comprehensive application. It is important for you to grasp basic knowledge and apply knowledge to analyze and solve problems.

- ✓ Attending classes
- ✓ Reading textbook
- ✓ Completing and turning in homework assignments in time



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SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

Room 406
TUE (every)
1-2

Room 406
THU (even)
1-2

Mid-term Exam

Final Exam

2024年	周次	一	二	三	四	五	六	日	国家节假日	重大活动	教学安排(本科)	教学安排(研究生)
9月	暑假	2 三十	3 八月	4 初二	5 初三	6 初四	7 白露	8 初六	9月17日 中秋节 (9月15日-17日 休息)	9月2日-9月8日 本科新生 入学教育 9月2日-9月6日 研究生新 生入学教育周 9月5日 本科生开学典礼 9月6日 研究生开学典礼 9月19日 全校教师大会暨 教师表彰大会 9月20日 纪律教育学习月 动员大会	9月6日 在校本科生返校 9月9日 本科生上课 9月14日 上双周周一的课 9月9日-27日 本科生秋季学期退补选课 暨导师指导周 (1-3周) 9月29日 上双周五的课	9月6日 在校研究生返校 9月9日 研究生上课 9月14日 上双周周一的课 9月21日 研究生导师培训会 9月29日 上双周五的课
	第1周 秋季学期	9 初七	10 初八	11 初九	12 初十	13 十一	14 十二	15 十三				
	第2周 秋季学期	16 十四	17 中秋节	18 十六	19 十七	20 十八	21 十九	22 秋分				
	第3周 秋季学期	23 廿一	24 廿二	25 廿三	26 廿四	27 廿五	28 廿六	29 廿七				
10月	第4周 秋季学期	30 廿八	1 国庆节	2 三十	3 九月	4 初二	5 初三	6 初四	10月1日 国庆节 (10月1日-7日休息)	10月1日 国庆升旗仪式 10月24日 本科招生工 作会议	10月12日 上双周周二的课	10月12日 上双周周二的课 10月-12月 2025级第一批博士 生招生录取
	第5周 秋季学期	7 初五	8 寒露	9 初七	10 初八	11 重阳节	12 初十	13 十一				
	第6周 秋季学期	14 十二	15 十三	16 十四	17 十五	18 十六	19 十七	20 十八				
	第7周 秋季学期	21 十九	22 二十	23 霜降	24 廿二	25 廿三	26 廿四	27 廿五				
11月	第8周 期中考试周	28 廿六	29 廿七	30 廿八	31 廿九	1 十月	2 初二	3 初三		11月5日 学术委员会2024年 第2次长聘与晋升评估会议 11月15日 2024年第四次校 学位会 11月15日-16日 第十届秋季 运动会 (15日停课一天) 11月16日 第四届南方科技 大学文化艺术节	10月28日-11月10日 本科生期中考试周(不停课)	10月28日-11月10日 研究生期中考试周 (不停课)
	第9周 期中考试周	4 初四	5 初五	6 初六	7 立冬	8 初八	9 初九	10 初十				
	第10周 秋季学期	11 十一	12 十二	13 十三	14 十四	15 十五	16 十六	17 十七				
	第11周 秋季学期	18 十八	19 十九	20 二十	21 廿一	22 小雪	23 廿三	24 廿四				
12月	第12周 秋季学期	25 廿五	26 廿六	27 廿七	28 廿八	29 廿九	30 三十	1 冬月		12月5-8日 重点中学校长 研讨会 12月7日 校友会第二届理 事会第四次会议 12月12日 2024年教职工 代表大会 12月14日 2024年度科研 工作会议 12月15日 校庆晚会 (迎 新年晚会暨南科好声音决赛) 12月18日 2024年第五次 校学位会 12月26日 2024年党组织书 记抓基层党建述职评议考核 12月28日 教育基金会第三		12月下旬 全国硕士研究生 招生考试
	第13周 秋季学期	2 初二	3 初三	4 初四	5 初五	6 大雪	7 初七	8 初八				
	第14周 秋季学期	9 初九	10 初十	11 十一	12 十二	13 十三	14 十四	15 十五				
	第15周 秋季学期	16 十六	17 十七	18 十八	19 十九	20 二十	21 冬至	22 廿二				
	第16周 秋季学期	23 廿三	24 廿四	25 廿五	26 廿六	27 廿七	28 廿八	29 廿九				

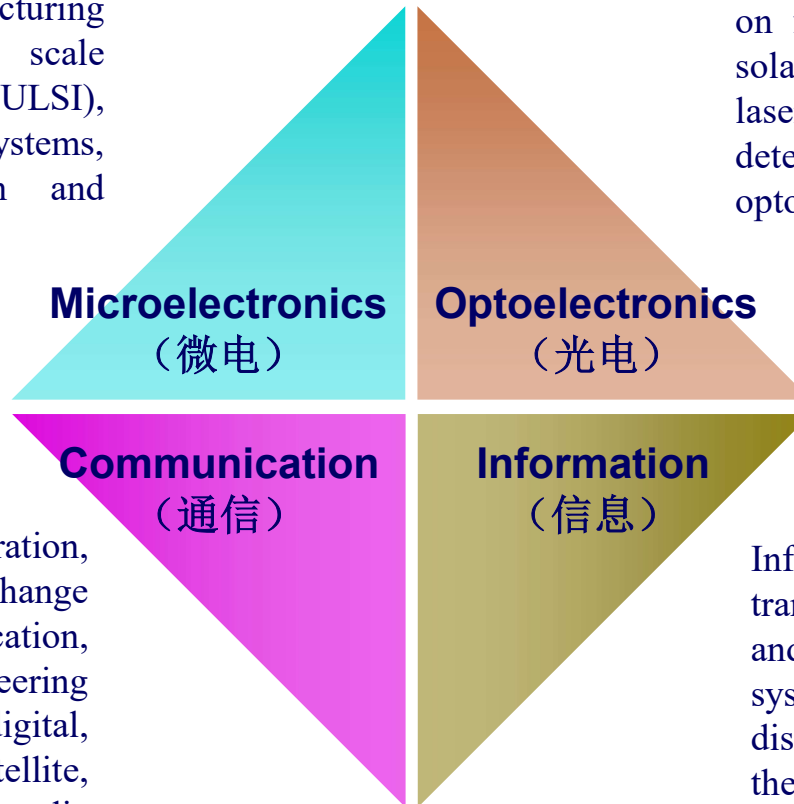


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Electronic Science and Technology

The research focuses on semiconductor devices, functional electronic materials, solid state devices, design and manufacturing technologies for ultra-large scale integrated circuits (ULSI), micromechanical electronic systems, and computer-aided design and manufacturing technologies.

Light radiation, transmission, detection and conversion, storage, processing and display, mainly focus on flat panel display, illumination, solar energy, optical communication, laser, optical design, photoelectric detection, semiconductor optoelectronic devices, etc.



Focus on signal generation, information transmission, exchange and processing in communication, and theoretical and engineering applications in wireless, digital, mobile, optical fiber, satellite, personal, multimedia communication, communication network, etc. problem.

Information acquisition, transmission, processing, utilization, and analysis and design of control systems. It is a specialized discipline that studies information theory, technology and engineering based on VLSI technology and modern computer technology.



Solid-State Electronics

半导体
材料与器件

Semiconductor
Materials and
Devices

Functional
Dielectric
Materials and
Devices

功能电介质
材料与器件

Electric Materials
& Devices

(电子材料与器件)

磁性
材料与器件

Magnetic
Materials and
Devices

Optoelectronic
Materials and
Devices

光电
材料与器件



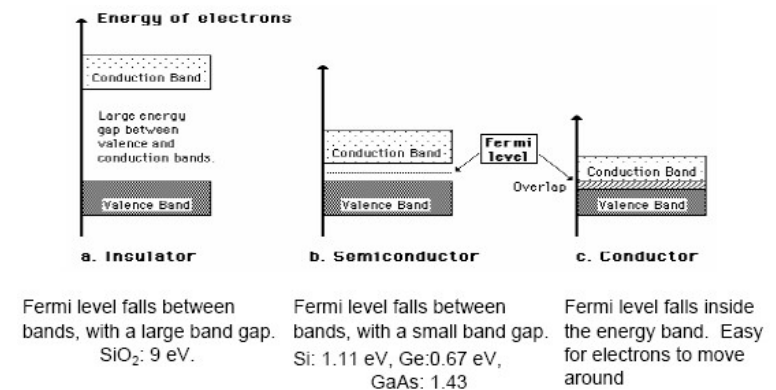
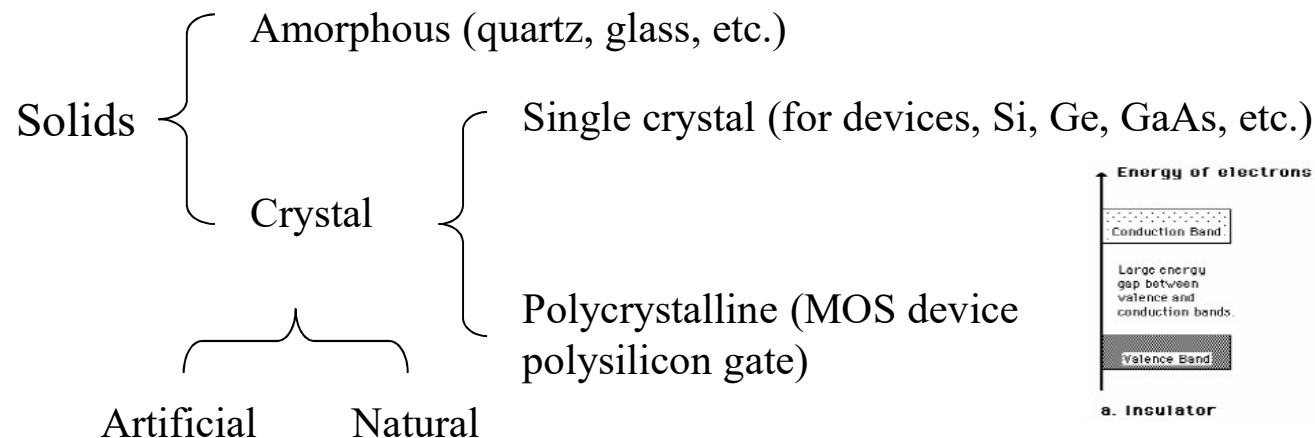
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Solid-State Electronics

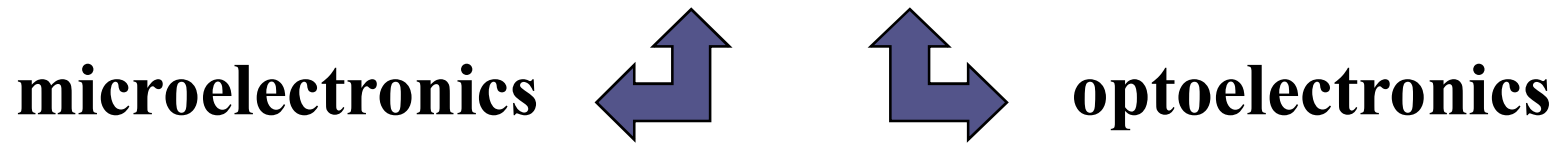
Solid-State Electronics 《固态电子学》 : Chapter 1-4

Introduction to Semiconductor Devices 《半导体器件导论》 : Chapters 5-9

Understand the laws of electron transport in solid at atomic scales; Understand macroscopic electrical and magnetic properties; Correctly distinguish conductors, semiconductors, and insulators; Correctly understand the characteristics of chemical bonds (electron binding methods and energy states); Understand the basic concepts of quantum mechanics, the generation of energy bands and the formation of energy levels.



Solid-State Electronics



Study the micro devices, circuits, and systems formed on solid (mainly semiconductor) materials; Study the generation, regulation, and application of photons under the interaction of light and matter.

Research Areas

- Light and matter interaction
- Semiconductor device physics
- Integrated circuit process
- Integrated circuit design and testing

Characteristics

- High integrated level, low power, high performance and high reliability circuits
- Infiltrate with other disciplines to form new subject areas: optoelectronic integration, MEMS, biochips



What is the core of the semiconductor industry?

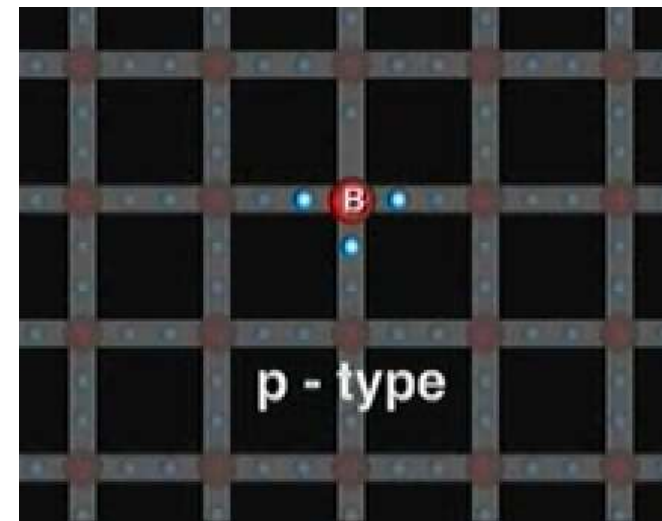
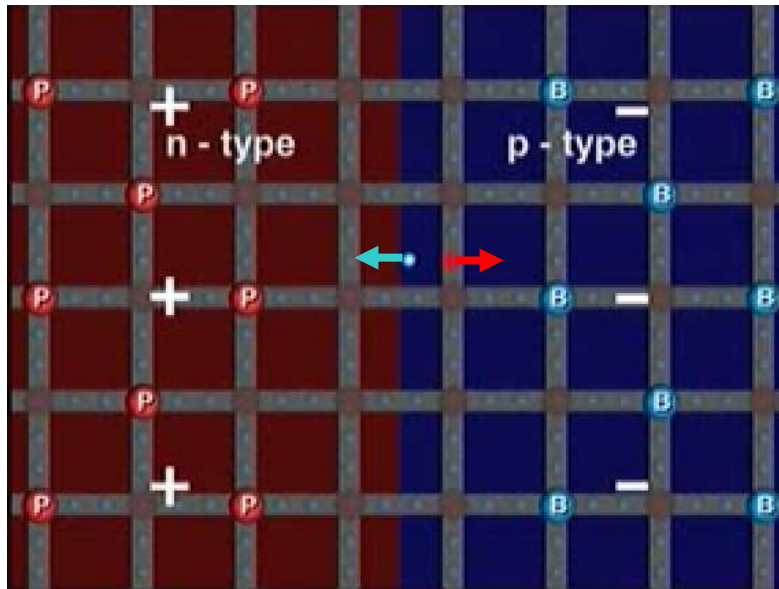
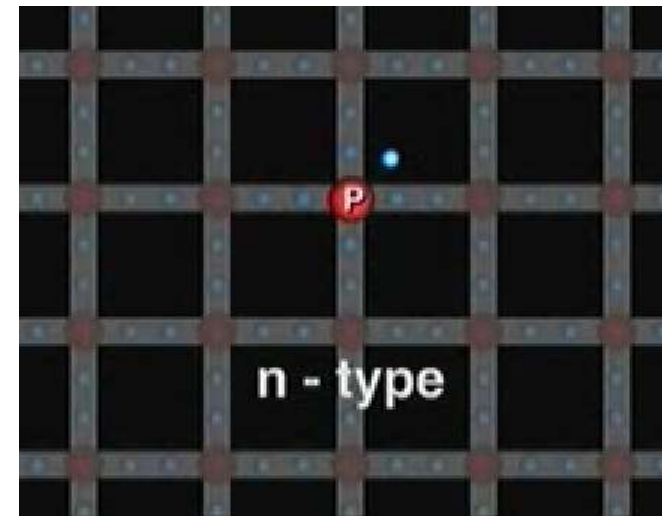
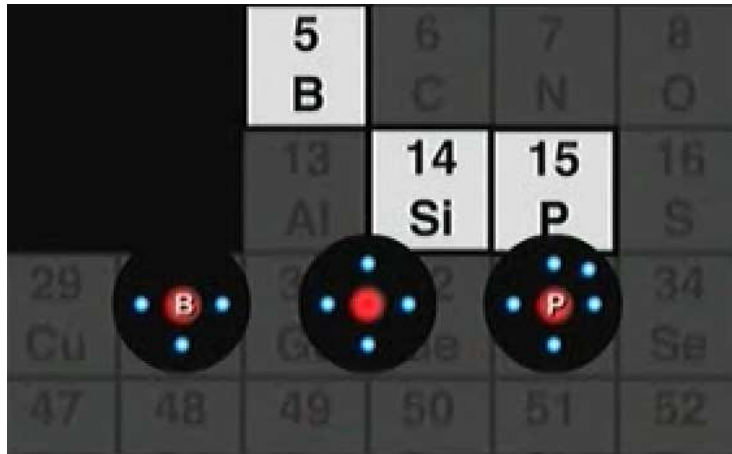
Electronics and semiconductor industry have surpassed the traditional steel industry and automobile industry and become a high value-added, high-tech industry in the 21st century. The rapid development of the electronics industry depends on the rapid improvement of the semiconductor industry. In the semiconductor industry, its core is integrated circuits (**electrical integration, optical integration, optoelectronic integration**). The rapid development of integrated circuits in terms of performance, integration, speed, etc. Based on the development of semiconductor physics, semiconductor devices, and semiconductor manufacturing processes.

Core: **集成电路** 【Integrated Circuit: IC】

Through a series of specific planar manufacturing processes, active devices such as transistors and diodes and passive components such as resistors and capacitors are "integrated" on a single semiconductor wafer and packaged in a protective case according to a certain circuit interconnection relationship. Within, a complex electronic system capable of performing specific functions.

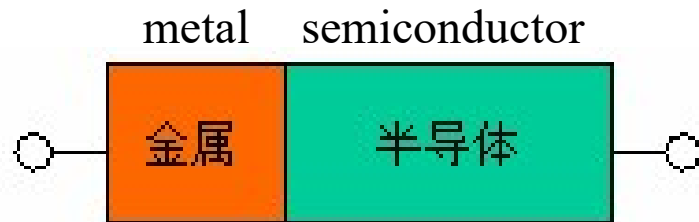


Doping of Semiconductor

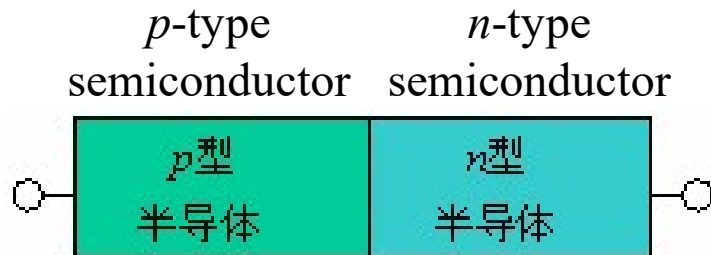


Basic Component of Devices

Humans have been researching semiconductor devices for more than 125 years, and so far there are about 60 major devices and 100 variant devices related to major devices. But all of these devices can be made up of several basic device structures.



Metal-semiconductors are used for ohmic and Schottky contacts (rectifying effects 整流效果); rectified contacts are used as gates, and ohmic contacts are used as drains and sources.

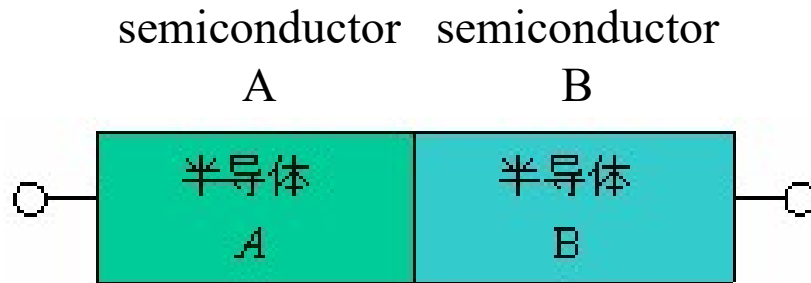


The p-n junction is a key component for semiconductor devices, and its theory is the basis of the physics of semiconductor devices. A *p-n-p* bipolar transistor can be formed to form a device of *p-n-p-n* structure.

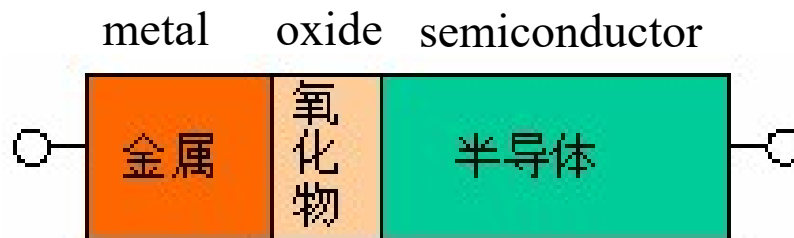


Basic Component of Devices

Humans have been researching semiconductor devices for more than 125 years, and so far there are about 60 major devices and 100 variant devices related to major devices. But all of these devices can be made up of several basic device structures.



Heterojunctions are a key component of fast devices and optoelectronic devices.

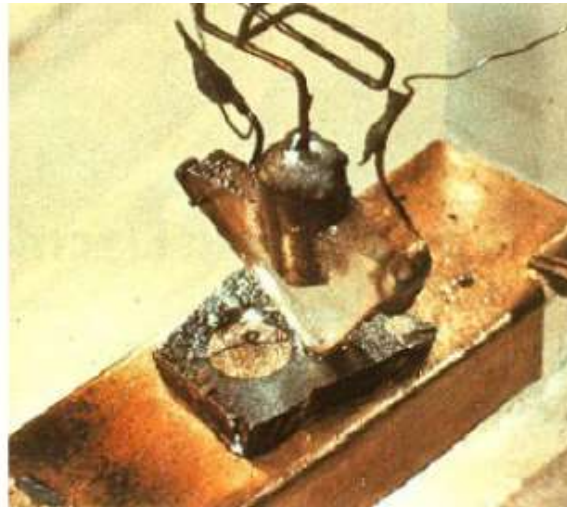
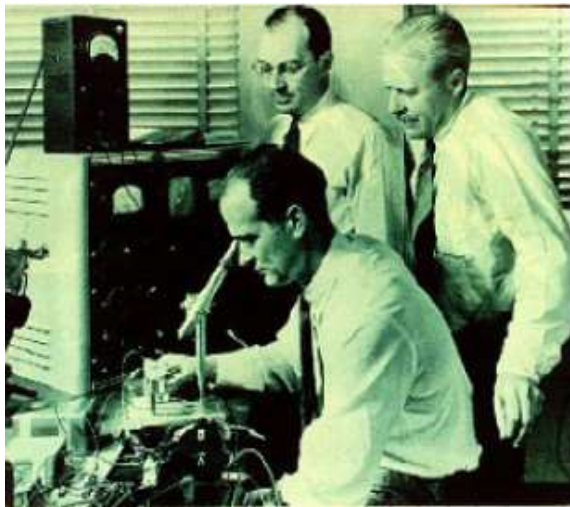


The MOFET can be fabricated by using the MOS structure as the gate and the two p-n junctions as the drain and source, respectively.



The Invention of Transistor

- In January 1946, Bell Labs officially established the Semiconductor Research Group, **W. Schokley**, J. Bardeen, and W. H. Brattain.
- Bardeen proposed the surface state theory, Schokley gave the basic idea of implementing the amplifier, and Brattain designed the experiment.
- On December 23, 1947, the transistor with amplification was first observed.



The Nobel Prize in Physics 1956

"for their researches on semiconductors and their discovery of the transistor effect"



William Bradford Shockley



John Bardeen

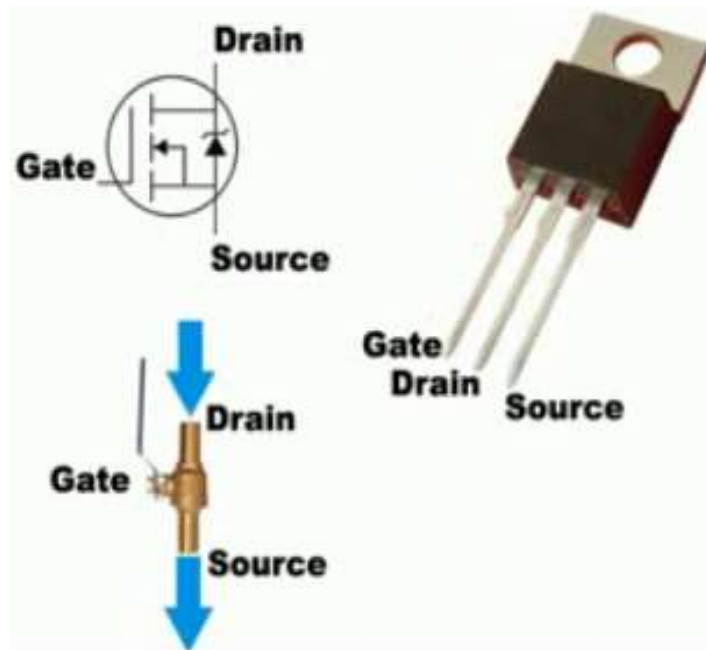
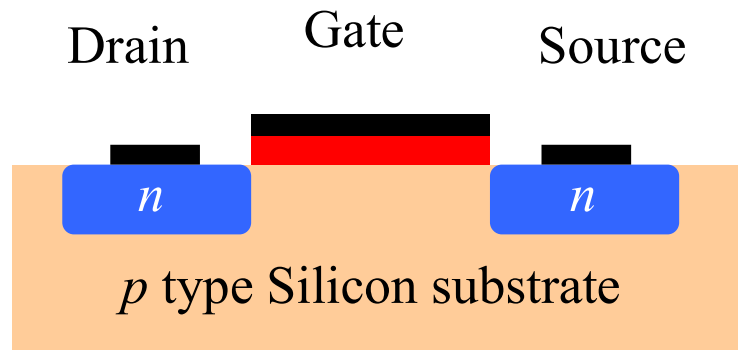


Walter Houser Brattain



南方科技大学
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The Invention of Transistor



**Transistor :
building block**



Fairchild Semiconductor
“仙童” 半导体

“八叛逆”
The Traitorous Eight



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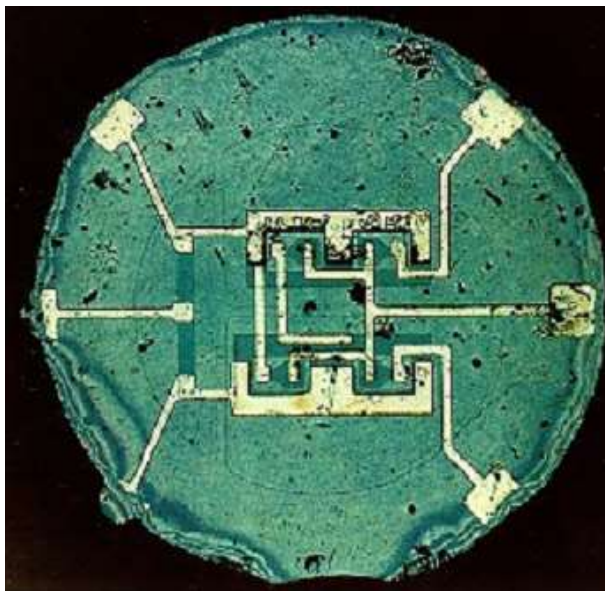
Various Semiconductor Devices

Year	Semiconductor Devices	Inventor
1874	金属-半导体接触 (MO)	Braun
1970	发光二极管 (LED)	Round
1947	双极型晶体管 (BJT)	Bardeen, Brattain and Shochley
1949	p-n结 (p-n Junction)	Shockley
1952	可控硅器件 (Thyristor)	Ebers
1954	太阳能电池 (Solar Cells)	Chapin, Fuller, and Pearson
1957	异质结双极型晶体管 (HBT)	Kroemer
1958	隧道二极管 (Tunnel diode)	Esaki
1960	金氧半场效应晶体管 (MOSFET)	Kahng and Atalla
1962	激光 (Laser)	Hall, et al.
1963	异质结激光 (Heterostructure Laser)	Kroemer, Alferov, and Kazarinov
1963	转移电子二极管 (TED)	Gunn
1965	碰撞电离雪崩渡越时间二极管 (IMPATT diode)	Johnston, Deloach, and Cohen
1966	金半场效应晶体管 (MESFET)	Mead
1967	非挥发性半导体存储器 (NVSM)	Kahng, and 施敏
1970	电荷耦合元件 (CCD)	Boyle and Smith
1974	共振隧道二极管 (Resonant-Tunneling Diode)	张立纲, Esaki, and Tsu
1980	调制掺杂场效应晶体管 (MODFET)	Mimura, et al.
1994	室温单电子存储器 (SEMC)	Yano, et al.
2001	15nm金氧半场效应晶体管 (15nm MOSFET)	Yu, et al.

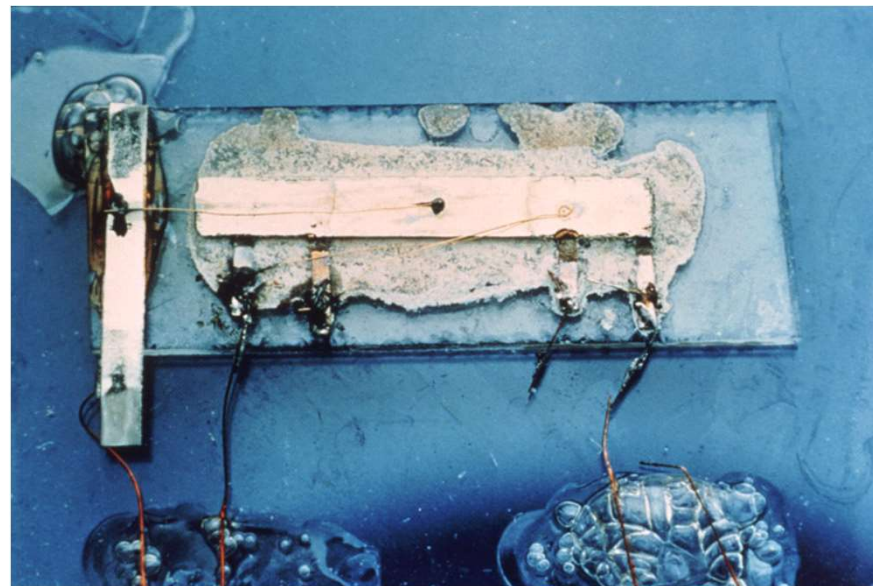


The Invention of Integrated Circuit

- In May 1952, British scientist G. W. A. Dummer first proposed the idea of an integrated circuit.
- In 1958, a research team led by TI's scientist Clair Kilby developed the world's first integrated circuit, and the results were published in 1959.
- Intel's Robert Noyce also invented the concept of IC's single crystal manufacturing.



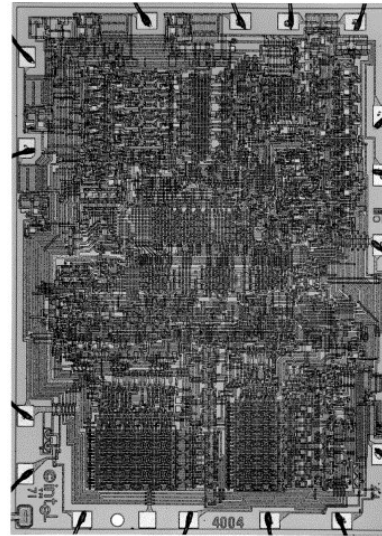
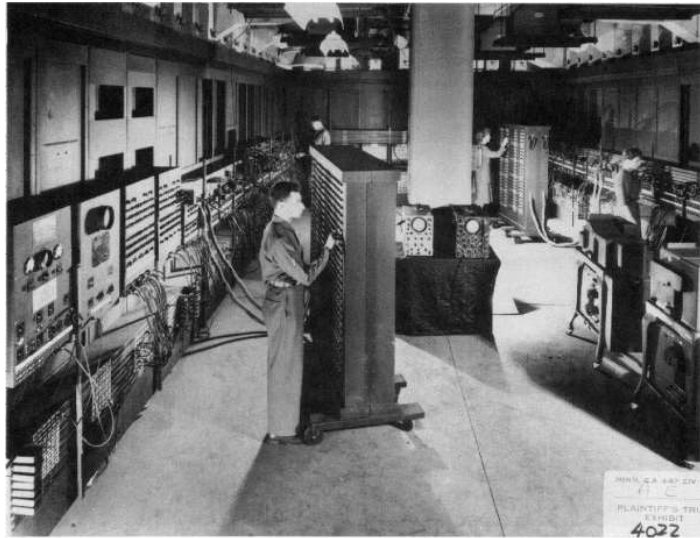
Robert Noyce(Intel)



Clair Kilby (TI)



ENIAC - The First Electronic Computer (1946)



Intel 4004 Micro-Processor

1971
1000 transistors
1 MHz operation



Domestic Processor
Mainland China

Godson No. 1 (Shenzhou Loondson Company)

Based on a 0.18 micron CMOS, 32-bit microprocessor. Support for the latest versions of Linux, VxWork, Windows CE and other operating systems. Can be widely used in industrial control, information appliances, communications, network equipment, PDA, network terminals, storage servers, security servers and other products.



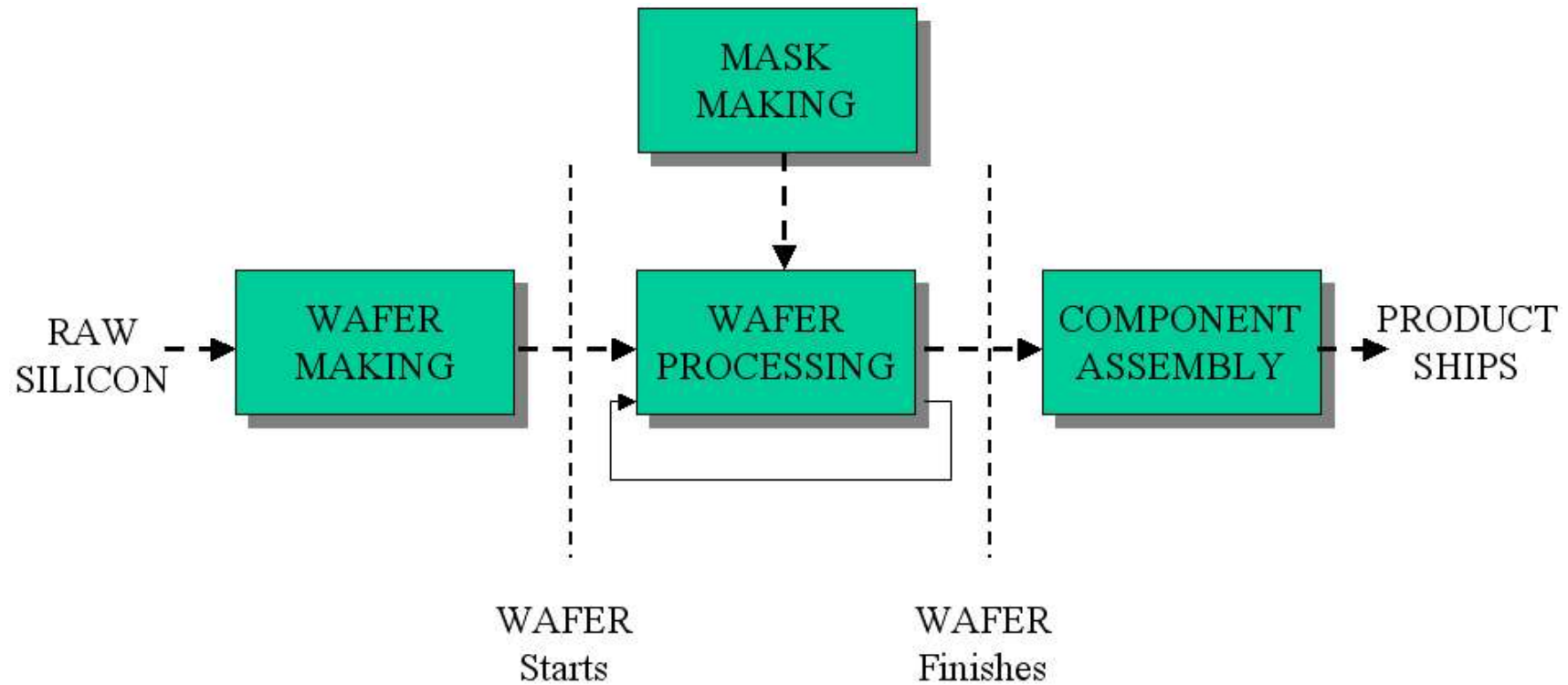
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Semiconductor Technologies

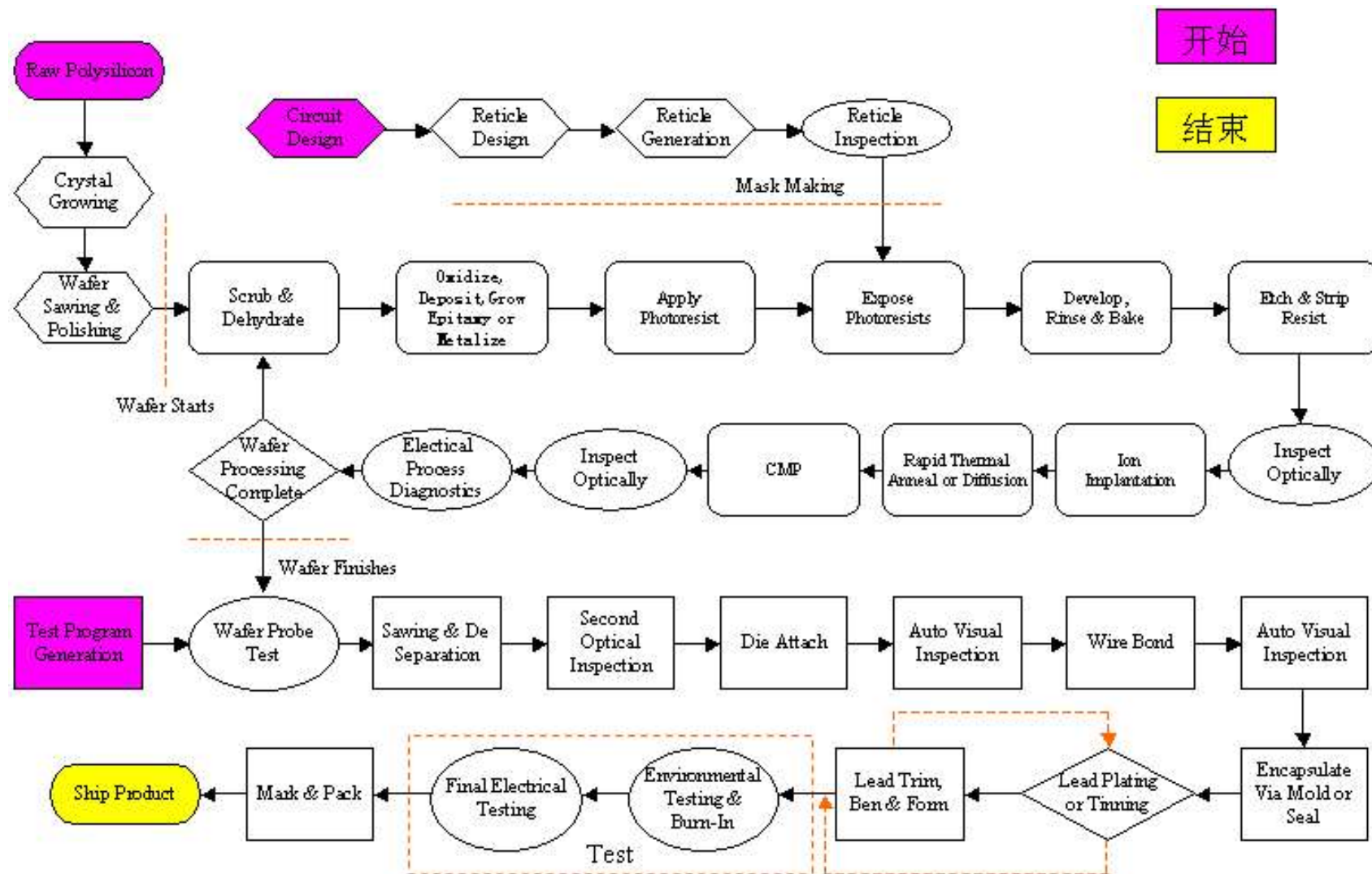
Year	Semiconductor Devices	Inventor
1918	柴可拉斯基晶体生长 (Zochralski)	Czochralski
1925	理吉曼晶体生长 (Lieman crystal growth)	Bridoman
1952	III-V族化合物 (III-V Group Semiconductor)	welker
1952	扩散 (Diffusion)	Pfann
1957	图形曝光抗蚀剂 (Exposure)	AndrMs
1957	氧化物掩蔽层 (Mask)	Frosch, and Derrick
1957	化学气相淀积 (Chemical vapor deposition: CVD)	Sheftal, Kokorlsh, and KrasiloV
1958	离子注入 (Ion implantation)	Shockley
1958	混合型集成电路 (Hybrid integrated circuit)	Kilby
1959	单片集成电路 (Monolithic integrated circuit)	Noyce
1960	平面化工艺 (Planarization process)	Hoerni
1963	互补式金氧半场效应晶体管 (CMOS)	Wanlass, and 萨支唐
1967	动态随机存储器 (DRAM)	Dennard
1969	多晶硅自对准栅极 (Polysilicon self-aligned gate)	Kerwin, Klein, and Sarace
1969	金属有机化学气相淀积 (MOCVD)	Manasevit, and Simoson
1971	干法刻蚀 (Dry etching)	Irving, Lemons, and Bobos
1971	分子束外延 (MBE)	卓以和
1971	微处理器 (4004)	Hoff, et al.
1982	沟槽隔离 (Gully isolation)	Rung, Momose, and Ogakubo
1989	化学机械抛光 (Mechanical polishing)	Davari, et al.
1993	铜布线 (Copper wiring)	Paraszcza, et al.



半导体芯片的制造框图

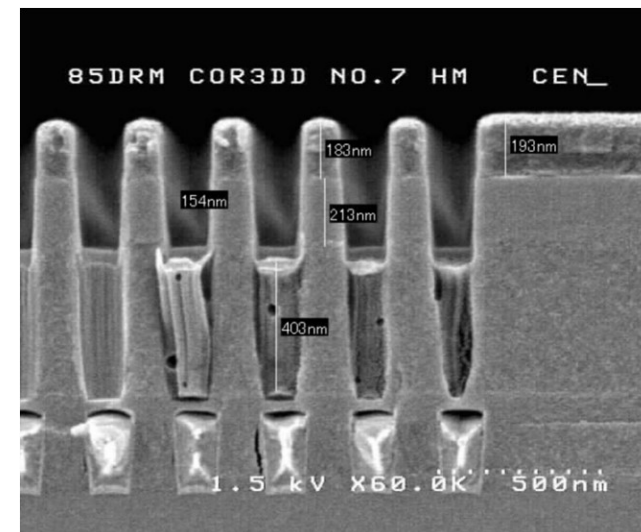
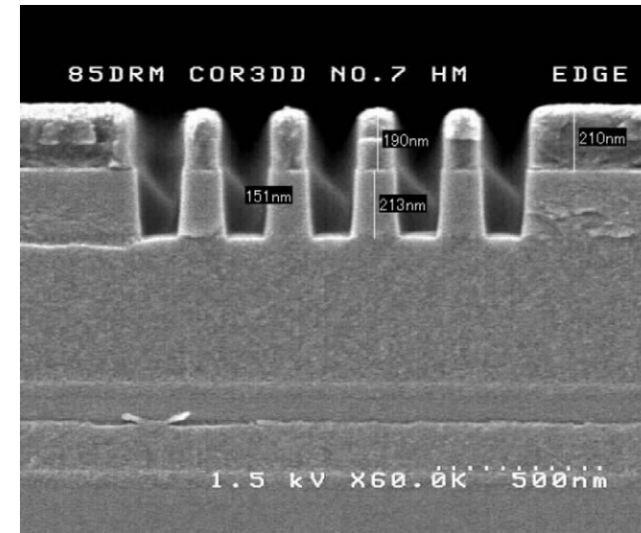
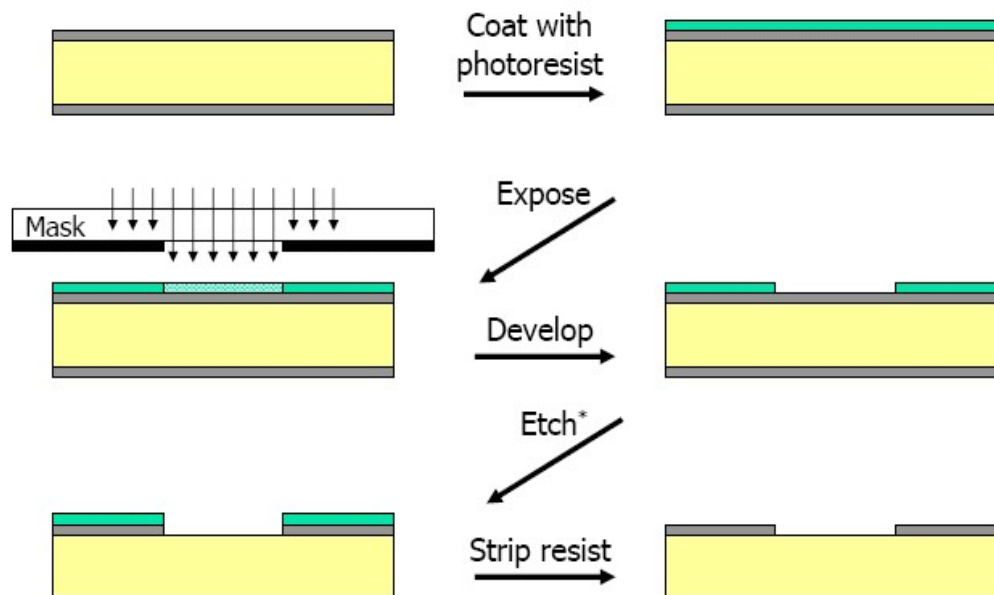


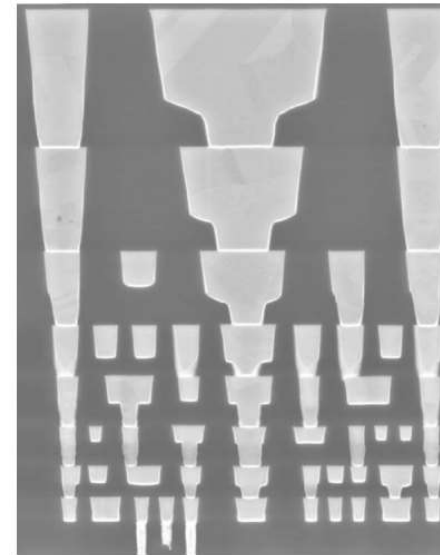
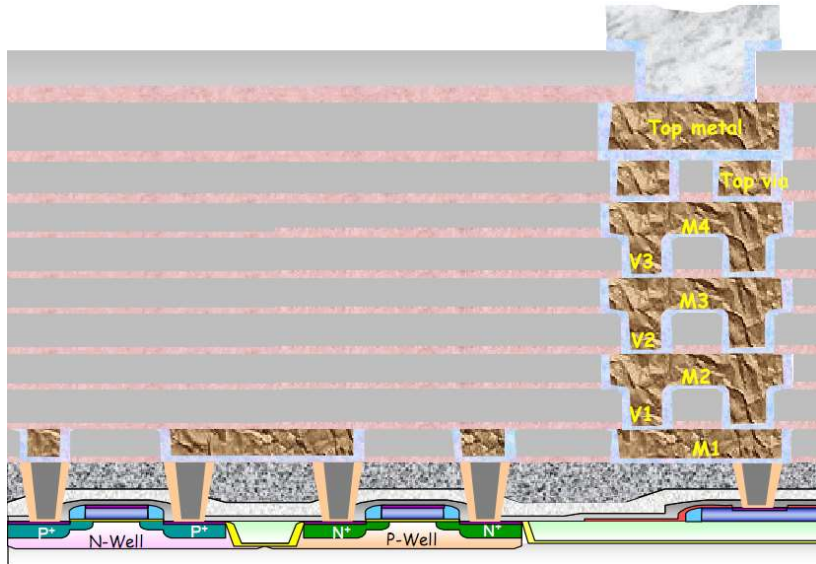
Typical Chip Manufacturing Process



Key Steps in Chip Manufacturing

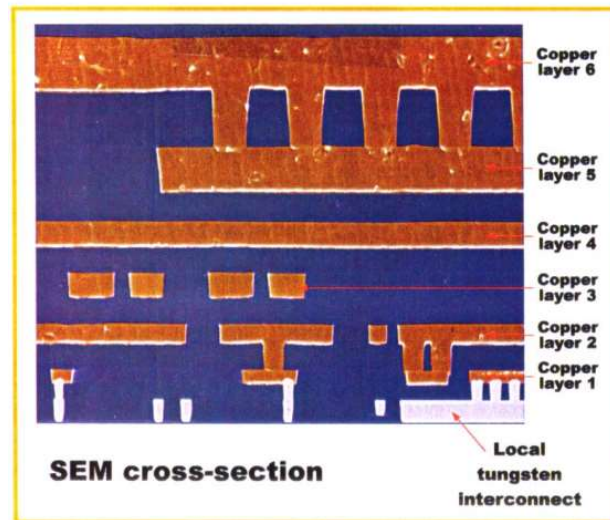
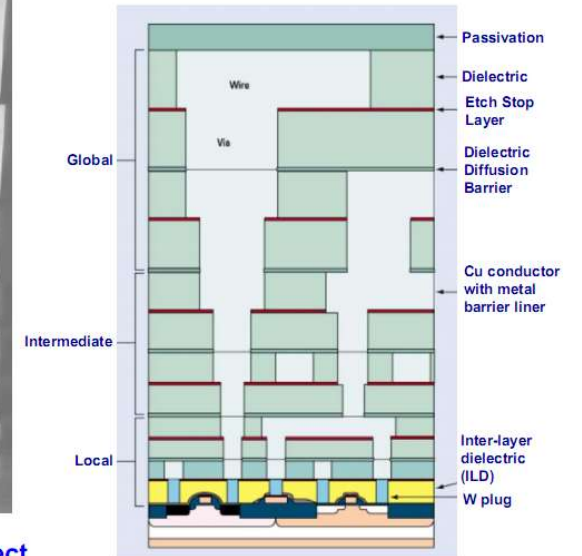
Lithography and Etching



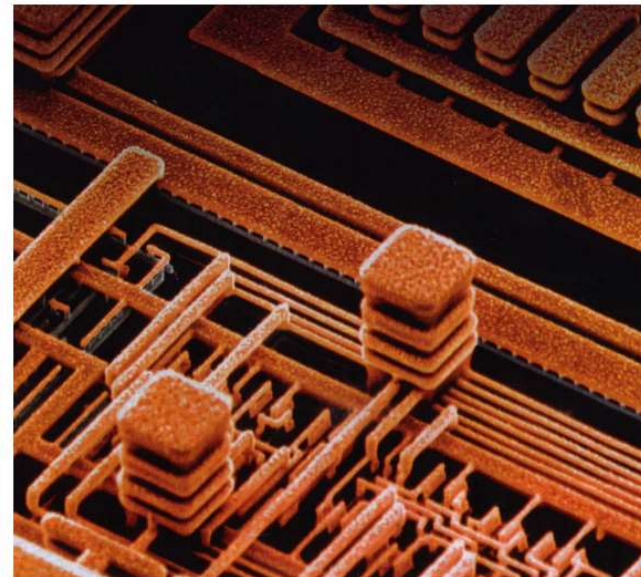


Intel's 65nm multi-level interconnect

Typical Chip Cross Section



■ Scanning electron micrograph illustrating IBM's new copper-based CMOS 7S technology. The process features six layers of copper metal lines and uses design rules that start at 0.2 μm . The 1.8-V technology is said to be capable of packing between 150 million and 200 million transistors on a chip.



Semiconductor Manufacturing

Semiconductor Manufacturing companies can be divided into 2 categories:

Design / Manufacturing Enterprise

Many companies have chip design and chip manufacturing, from the front-end design of the chip to the back-end processing. Intel, IBM, Motorola, Samsung, Hynix, Infineon, Philips, STmicroelectronics, etc.

Foundry Company

In the chip manufacturing industry, there is a special type of enterprise that specializes in manufacturing chips for other chip design companies. These companies are called foundry. The emergence of foundry is due to the rapid development of modern technology, more and more technology requires a more detailed division of labor, which can partially reduce the cost or risk of the enterprise.

For example, graphics cards and motherboards, whose core is graphics processors and chipsets, are designed by some top-level chip development companies like nVIDIA, ATI, INTEL, AMD, VIA, SIS, ALI, etc., and then commissioned to some factories to process them into Chips and chipsets.



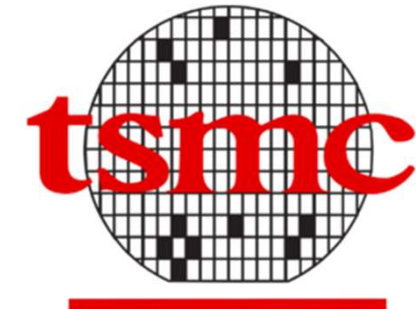
Famous Foundries

台积电 (TSMC)

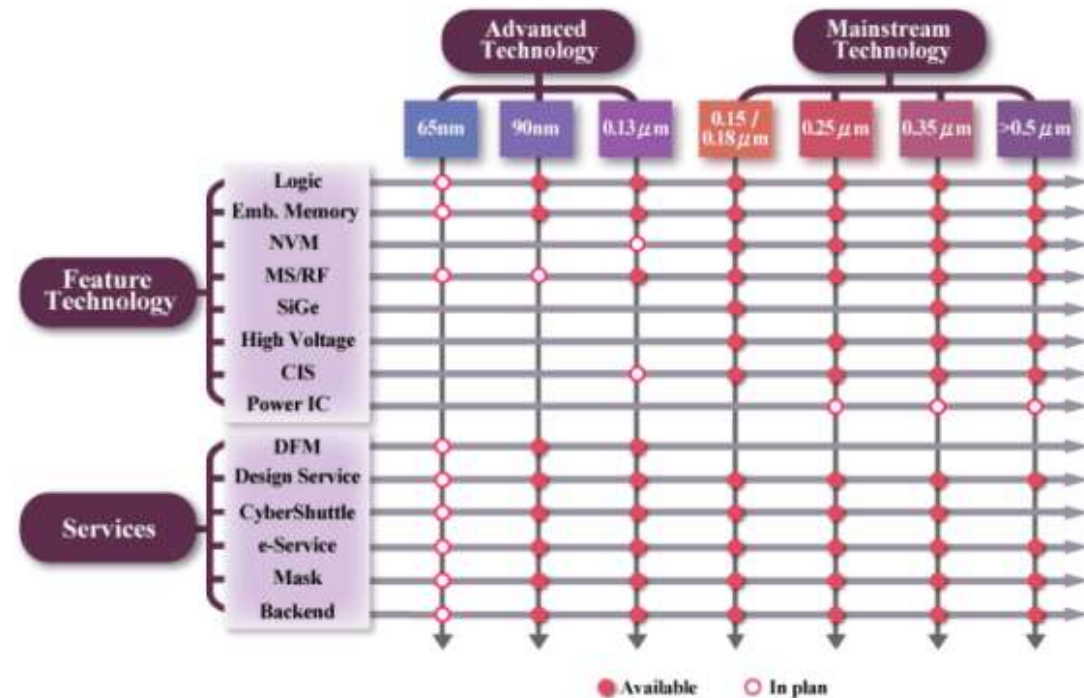
The graphics processing chips designed by ATI and nVIDIA, or the motherboards designed by VIA, SIS, and ALI are basically produced by TSMC and UMC. TSMC was founded by Mr. Zhang Zhongmou, the “Semiconductor Godfather of Taiwan”



张忠谋



台灣積體電路製造股份有限公司
Taiwan Semiconductor Manufacturing Company, Ltd.



南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY

Famous Foundries

台联电（UMC）：In 1980, Taiwan first integrated circuit company. Under the leadership of Cao Xingcheng, UMC has become the second largest semiconductor company in Taiwan after TSMC and the second largest professional chip foundry in the world.

TECHNOLOGY	VOLTAGE OPTIONS	MAX. METAL LAYERS
90 nm Copper Logic (L90)	1.0V/1.2V/3.3V, 1.0V/1.2V/2.5V 1.0V/1.2V/1.8V	9 Cu
0.13 um Copper Logic (L130)	1.2V/3.3V, 1.2V/2.5V 1.0V/3.3V, 1.0V/1.8V	8 Cu
0.13 um Copper Mixed Mode	1.2V/3.3V	8 Cu
0.15 um Logic (L150)	1.5V/3.3V, 1.2V/1.8V, 1.2V/3.3V	7
0.18 um Logic (L180)	1.8V, 1.8V/3.3V	6
0.18 um Copper Logic	1.8V/3.3V	4 Al, 2Cu
0.18 um Mixed Mode (RFCMOS)	1.8V/3.3V	6
0.25 um Logic (L250)	2.5V, 2.5V/3.3V, 2.5V/5V	5
0.25 um Mixed Mode	2.5V, 2.5V/3.3V, 2.5V/5V	5
0.25 um CMOS Image Sensor	2.5V/3.3V	5



曹兴诚

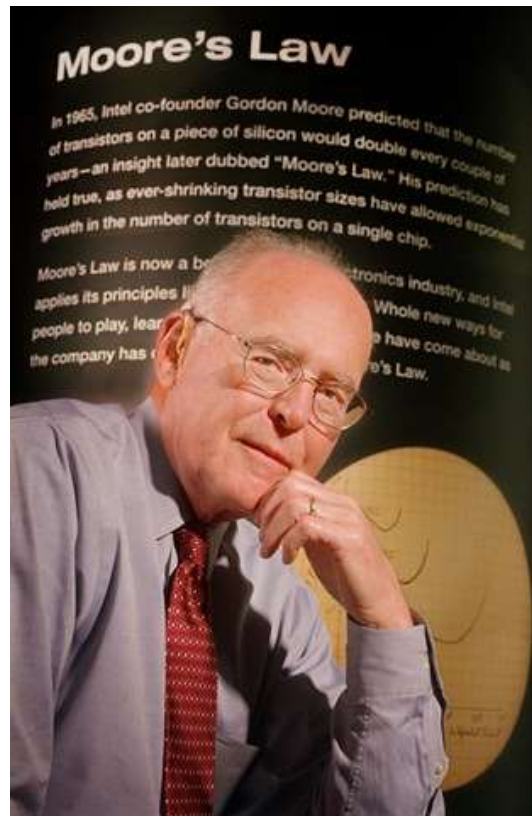


Famous Foundries

中芯国际 (SMIC) : was founded in 2000. The company is located in Shanghai, China, and has three chip foundries, including a back-end copper process foundry. Technical capabilities include logic circuits, mixed-signal/RF circuits, high-voltage circuits, system-on-chip, embedded and other memories, silicon-based liquid crystals, and image sensors.



Moore's Law

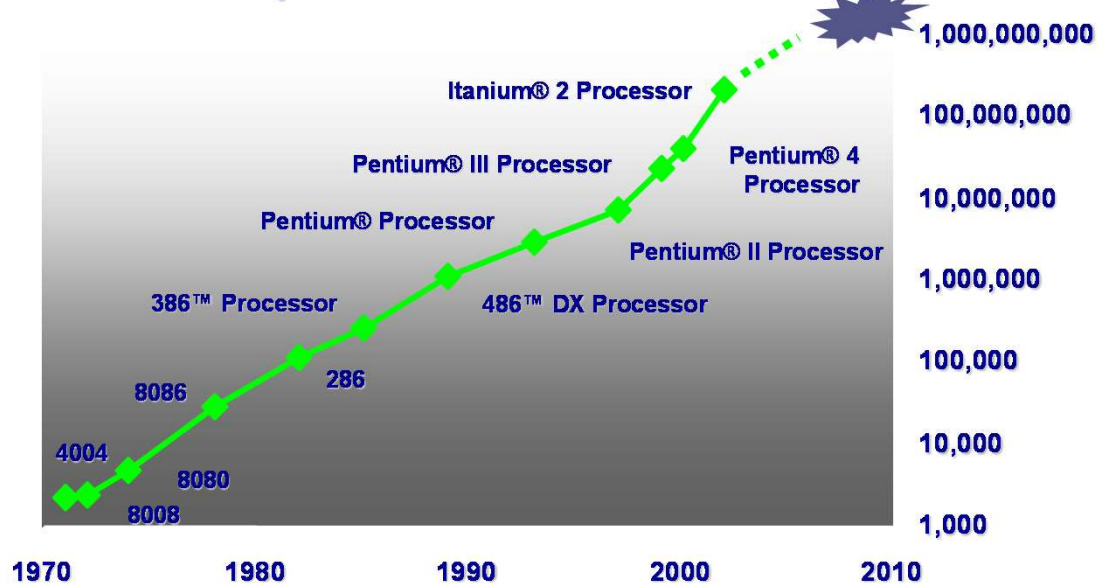


Intel cofounder Gordon Moore projected that the number of transistors on a computer chip will double every 2 years. (1965)

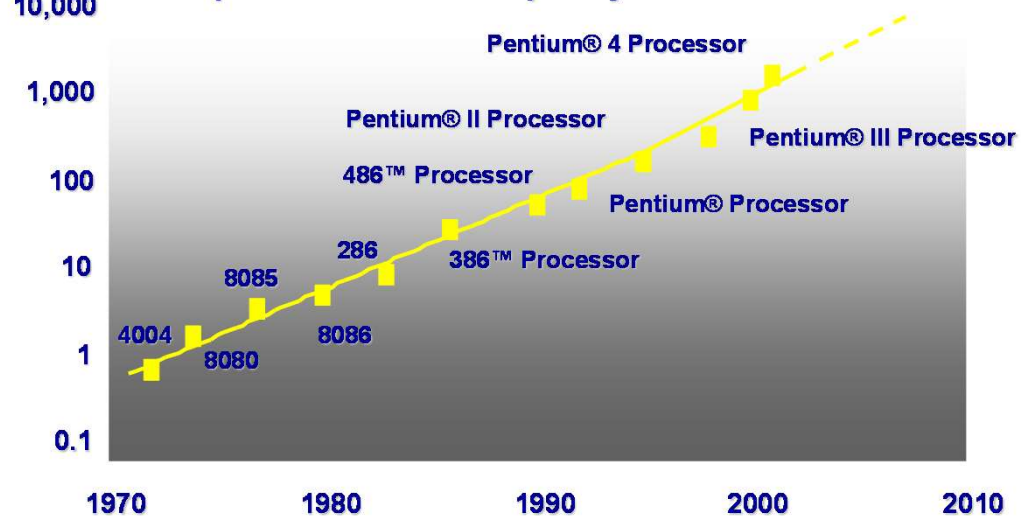


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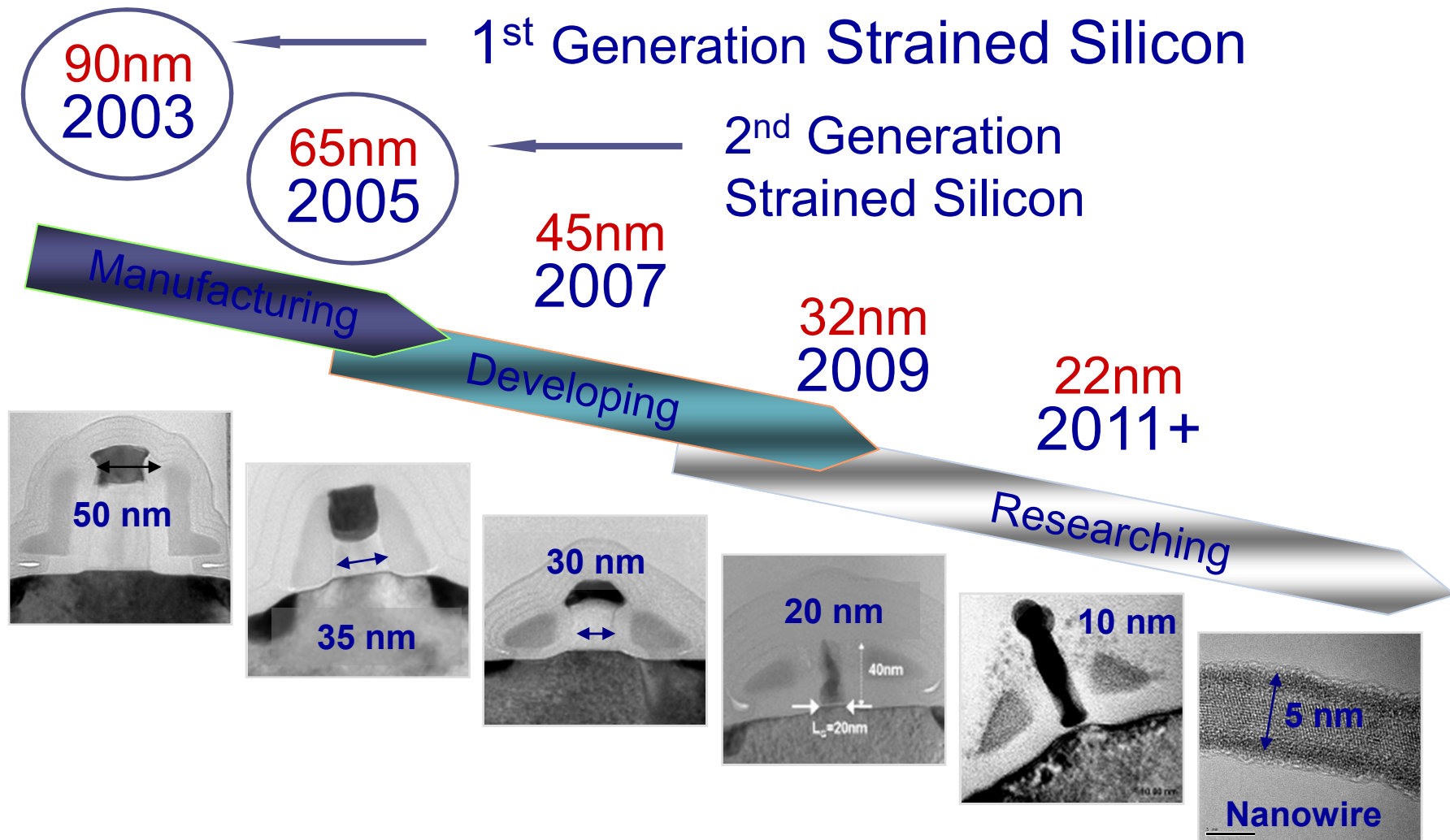
Microprocessor transistor count



Microprocessor Clock Frequency



Development Trend



Communication Applications

